

Database Triggers (Self-Learning)

Q. What is a Trigger in SQL? Discuss the different types of triggers (BEFORE and AFTER) and explain their primary uses.

📌 Definition to Remember

A **Trigger** is a special stored procedure in a database that **automatically fires (executes)** in response to a DML event (**INSERT** , **UPDATE** , or **DELETE**) on a specific table. Unlike regular procedures, **triggers are never called explicitly** — the RDBMS engine activates them automatically.

1. Types of Triggers

Triggers are classified by **when** they execute relative to the triggering DML event:

A. BEFORE Trigger

- Executes **immediately before** the INSERT, UPDATE, or DELETE completes.
- Purpose:** Validates or modifies incoming data before it is saved.
- Use Cases:** Data validation, enforcing complex constraints, transforming input data.

Example: Before inserting into the **Employee** table, a BEFORE trigger checks if the incoming salary is below minimum wage. If so, it **raises an error and aborts** the insert.

B. AFTER Trigger

- Executes **immediately after** the INSERT, UPDATE, or DELETE has successfully completed.
- Purpose:** Reacts to committed changes — used for auditing and cascading operations.
- Use Cases:** Audit logging, cascading updates to related tables, sending notifications.

Example: After an update to **Employee** , an AFTER trigger inserts a record into **Employee_Audit** logging the user, timestamp, and what changed.

(Some systems also support INSTEAD OF Triggers — primarily used on Views to enable update operations on non-updatable views.)

2. Trigger Event Classification

	BEFORE	AFTER
Timing	Before data is saved	After data is saved
Purpose	Validate / modify input	React / audit / cascade
Can Abort?	Yes — can raise error to stop	No — data already committed

3. Uses and Advantages

Use Case	Explanation
Business Rule Enforcement	Complex rules that CHECK constraints can't handle
Auditing	Auto-log who changed what and when
Task Automation	Calculate derived columns, update balances automatically
Referential Integrity	Enforce complex FK rules beyond standard constraints

★ Must-Write Points (for 10 marks)

1. A Trigger automatically fires on a DML event (INSERT, UPDATE, DELETE) — never called explicitly.
2. **BEFORE Trigger**: fires before the DML; used for data validation and input modification.
3. **AFTER Trigger**: fires after the DML; used for auditing and cascading operations.
4. BEFORE triggers can abort the operation by raising an error; AFTER triggers cannot.
5. Key uses: business rule enforcement, audit logging, task automation, referential integrity.
6. INSTEAD OF triggers are used on Views to allow updates on non-updatable views.
7. Poorly designed triggers can cause cascading errors and severe performance degradation.

⚡ Quick Recall

Trigger → Auto-fires on INSERT/UPDATE/DELETE → BEFORE (validate/abort) → AFTER (audit/cascade) → Uses: Business Rules, Auditing, Automation → Never explicitly called